Bioreactor Mixing: Liquid Mixing (Blending) and Gas-Liquid Dispersion

⇨ **Mechanical design**

⇨ **Power consumption**

⇨ **Oxygen transfer**

⇨ **Bioreactor mixing (Part 1)**

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General Concepts

- Mechanical agitation of a STR promotes various types of mixing: **liquid mixing or blending** (e.g. during fed-batch operation, pH adjustment etc), solid-liquid suspension, gas-liquid dispersion and liquid-liquid mixing
- Good mixing is necessary to ensure efficient mass and heat transfer and effective fermentation optimisation

Turbulent flow ≠ well mixed

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Liquid Mixing Mechanisms in Various (Single-Use) Bioreactor Geometries



Shaken



Rocked



Stirred

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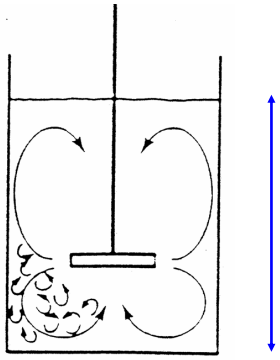
Radial and Axial Dispersion in a STR and Energy Dissipation by Turbulent Eddies



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Radial and Axial Dispersion in a STR and Energy Dissipation by Turbulent Eddies



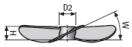
Axial mixing (dispersion)

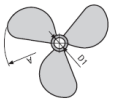
Radial mixing (dispersion)

6

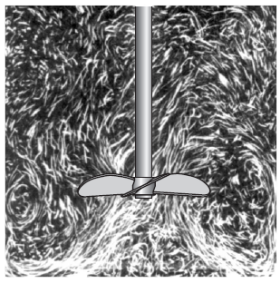
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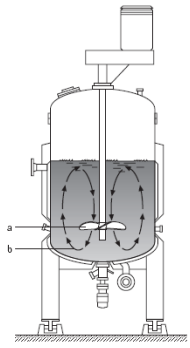
Influence of Impeller Design on Bioreactor Mixing: Marine Impeller









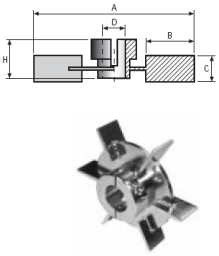


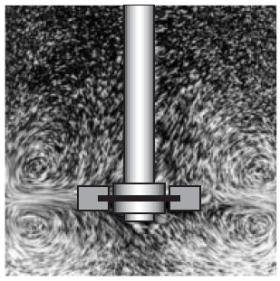
⇨ Low shear
⇨ Good liquid mixing
⇨ Low power number

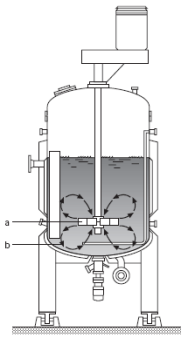
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Influence of Impeller Design on Bioreactor Mixing: Rushton Turbine Impeller







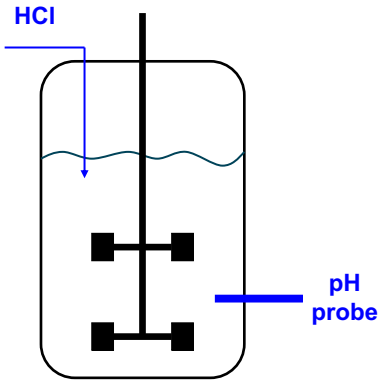
⇨ High shear
⇨ Good gas dispersion
⇨ High power number

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Mixing Time Definition and Measurement

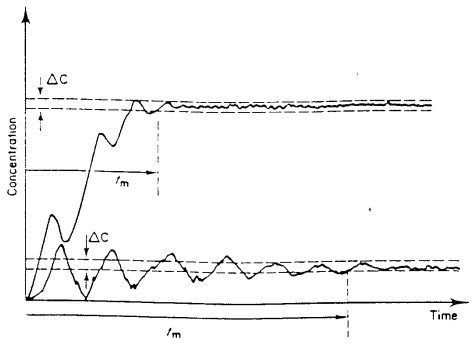
Mixing time, t_m, is the minimum time after introducing a tracer at which the concentration fluctuations lie within a predetermined concentration range (ΔC):



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Mixing Time Definition and Measurement

Mixing time, t_m, is the minimum time after introducing a tracer at which the concentration fluctuations lie within a predetermined concentration range (ΔC):



Circulation time, t_C. In turbulent regime:

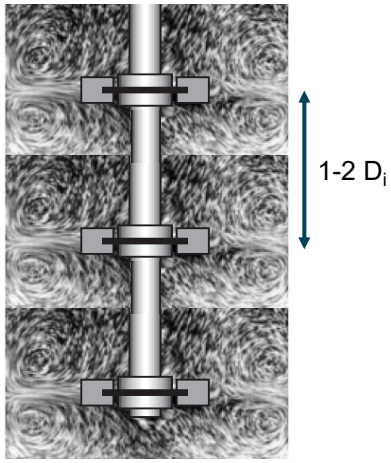
$t_m \sim 5\ t_c$

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Factors Influencing Mixing Time

- Agitator type
  (axial or radial dispersion)
- Number and position of agitators
  (compartmentalisation)
- Presence of baffles
- Location of component addition

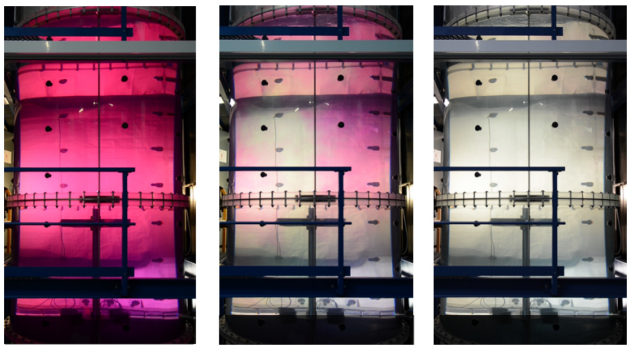
Compartmentalisation



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Compartmentalisation in a Large 15,000L Cell Culture Reactor (Rosseburg *et al*, Chem Eng Sci, 2018)



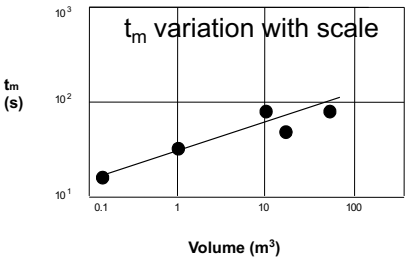


t_0 $0.5\ t_m$ t_m

(~ 60 sec)

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Significance of Mixing Time Measurements and Industrial Bioreactor Mixing



- If t_m is less than the smallest relevant time constant by an order of magnitude then mixing will not be a problem
- For oxygen transfer, if $k_La = 0.16\ s^{-1}$, then the time constant for oxygen transfer, $1/k_La = 6.3s$. In this case t_m values > 1 min will likely cause problems i.e. mixing becomes rate limiting and could lead to DOT variation within the reactor
- What impact will poor mixing have on glucose addition, oxygen transfer, fermenter control?

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Bioreactor Mixing: Liquid Mixing (Blending) and Gas-Liquid Dispersion

⇨ **Mechanical design**

⇨ **Power consumption**

⇨ **Oxygen transfer**

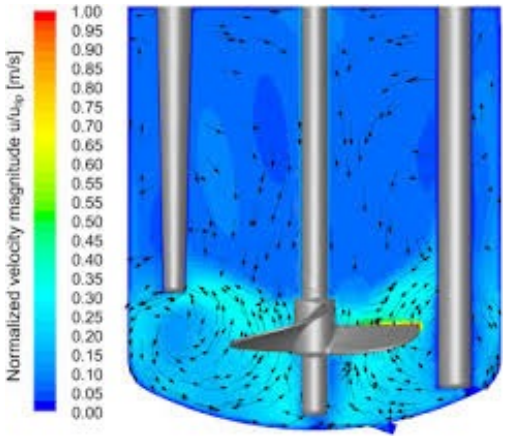
⇨ **Bioreactor mixing (Part 2)**

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Advanced Methods for Fluid Characterisation: Computational Fluid Dynamics

CFD simulation of fluid flow and velocity distribution in a **3L single-use stirred bioreactor** for mammalian cell culture.

Single marine-type impeller in an unbaffled tank with temperature and dissolved oxygen probes. Velocity values normalised to V_{tip}

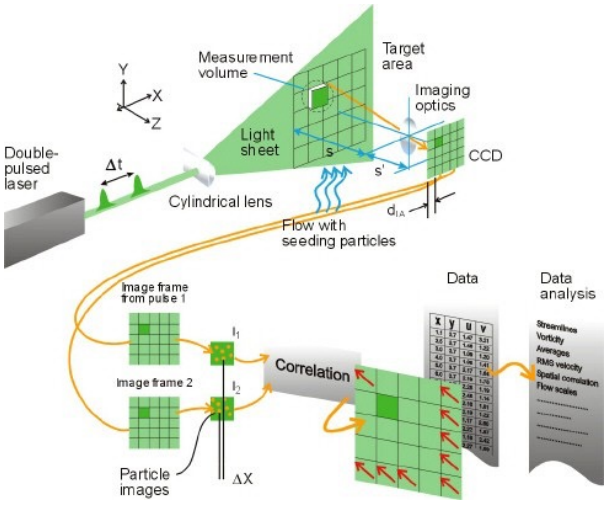


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Advanced Methods for Fluid Characterisation: Particle Image Velocimetry (PIV)

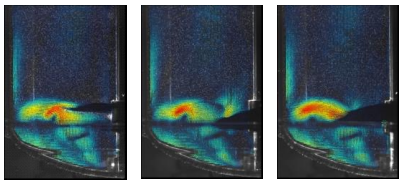
Picture from Dantec Dynamics Ltd website



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Advanced Methods for Fluid Characterisation: Particle Image Velocimetry (PIV)





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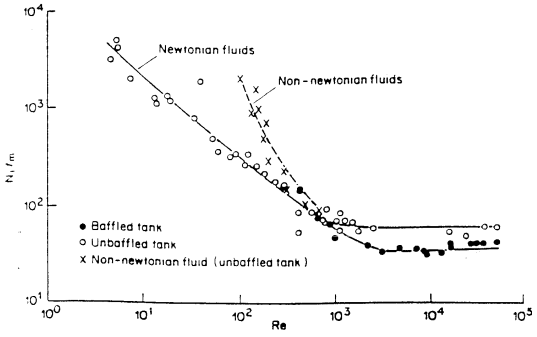
Mixing Time Correlations: Single Phase Fluids

In **turbulent regime**:

$N.t_m$ = constant

Geometry can also be important:

$N.t_m.(D_i/D_t)^a$ = constant



Other **empirical correlations** exist e.g.

$$\Theta_{99\%} = \frac{6.34}{N\left(\frac{D}{T}\right)^{2.3}\left(\frac{Z}{T}\right)^{-0.5}}$$

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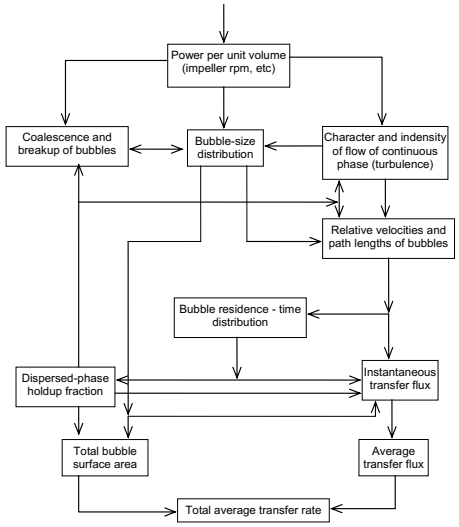
Effect of Gassing on Liquid Mixing

Mixing time in a gas-liquid system is typically taken as double the single phase value

Author	V (m^3)	P (W)	T (m)	t_m measured aerated	t_m calc single phase
Einsele (1978)	0.14	2100	0.45	15	6
	1	5500	0.86	42	12
	10	25000	1.85	84	26
	20	23000	2.34	51	38
	57	116000	3.31	82	40
Jansen *et al* (1980)	1.9	3800	1.08	29	24
	60	120000	3.45	67	52

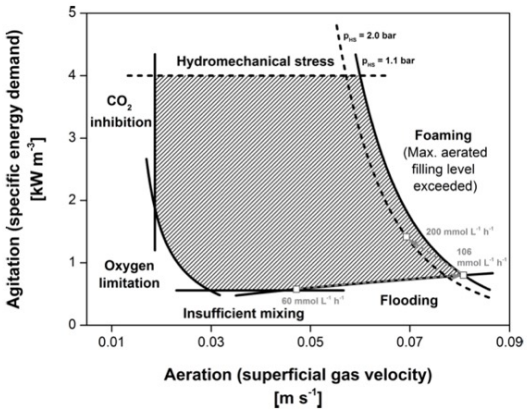
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Interaction of Bioreactor Geometry, Power Input, Mixing and Oxygen Transfer



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Interaction of Bioreactor Geometry, Power Input, Mixing and Oxygen Transfer



Operating Window for a large scale (100 m^3), aerated stirred bioreactor for bacterial culture

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